DENG HAOPENG

Guangzhou, China | +86 198-6670-0543 | qianyhp@gmail.com | Google Scholar | Homepage | GitHub

EDUCATION

Guangzhou Maritime University **09/2022-07/2026**

B. Eng. in Engineering Management, School of Future Transportation

Average Score: 83.86 Rank: Top 22% (32/145)

RESEARCH INTERESTS

Intelligent Transportation System, Mixed Traffic Dynamics, Multi-agent Control, Dynamic Traffic Assignment, Spatiotemporal Data Analysis, Urban Mobility Modelling, Optimization Algorithms

PUBLICATIONS

- **Deng, H.**, Zheng, F., Ma, K., et al. (2025). Adaptive Truncated Schatten Norm for Traffic Data Imputation with Complex Missing Patterns. *The 25th COTA International Conference of Transportation Professionals*. DOI: 10.21203/rs.3.rs-6804022/v1.
- **Deng, H.**, Zheng, F., & Zhan, L. (2024). Application of Markov Processes in Traffic Signal Control. *International Core Journal of Engineering*. DOI: 10.6919/ICJE.202411_10(11).0015.
- Wang, Y., Zhang, H., Zhou, Y., **Deng, H.**, et al. (2025). Exploring the 2-Part of Class Groups in Quadratic Fields: Perspectives on the Cohen–Lenstra Conjectures. *Mathematics.* DOI: 10.3390/math13010051.

MANUSCRIPTS

- Xia, X., **Deng, H.**, Tang, J., et al. (2025). A Review of Dynamic Traffic Assignment in Mixed Traffic: Modeling, Evolution, and Solution Approaches. Under review at *Journal of Transportation Engineering and Information.*
- Wang, Y., **Deng, H.**, Zhu, L., et al. (2025). Approximate Sparse Stochastic Control for Time-Varying Systems with Control-Dependent Diffusion. Under review at *Kybernetika*. Preprint DOI: 10.21203/rs.3.rs-6809362/v1.

RESEARCH EXPERIENCES

Guangdong Provincial Special Fund for Science and Technology Innovation Strategy **07/2024-04/2026**

Title: *Cooperative Game-Based Network Traffic Assignment Mechanisms for Enhancing Urban Transportation Efficiency* | *Grant No. PDJH2024A289* | *Project Leader*

- Authored a systematic review on dynamic traffic assignment (DTA), synthesizing insights from 100+ studies on multi-class vehicle modeling, nonlinear equilibrium evolution, and solution algorithms.
- Developed a VI-based DTA model that unifies UE–SO equilibria into a tunable spectrum, by decoupling CAV penetration into automation levels (AL) and connectivity levels (CL) as evolution variables.
- Proposed the AMPAS algorithm for solving DUE, DSO, and UE–SO mixed equilibria within the AL–CL framework, achieving 27.2% faster convergence than gradient projection on medium-scale networks.
- Direct team task delegation, milestone tracking, and mentoring of lower year members; coordinate regular progress meetings to align efforts and ensure on-schedule research deliverables.

Guangdong Provincial College Student Innovation & Entrepreneur Training Program **11/2024-02/2026**

Title: *Intelligent Recognition of Urban Traffic Equilibrium States under Air-Ground Collaborative Monitoring* | *Core Member (Second Rank)* | *Instructor: Prof. Xinhai Xia*

- Integrating heterogeneous data sources from roadside sensors, probe vehicles, and UAVs to construct high-dimensional traffic flow tensors, supporting accurate traffic equilibrium state recognition.
- Formulated three representative real-world data loss structures—element-wise, fiber-wise, and mixed—within urban traffic flow tensors, serving as a rigorous testbed for refining data recovery algorithms.
- Developed an adaptive truncated Schatten norm-based low-rank tensor completion model to effectively fill gaps in traffic datasets with complex missing patterns, improving MAPE by 10.6% and RMSE by 6.1%.
- Facilitated industry-academia partnership for traffic survey equipment and training, guided junior students in fieldwork to create datasets for analysing traffic flow dynamics and defining equilibrium states.

The Hundred Thousand Talent Plan **07/2024**

Title: *Investigation and Research on Traffic Safety and Planning in Kaiping City | Project Leader*

- Collaborated with the Municipal Transportation Department to analyse traffic accident and violation data, identifying accident hotspots and associated hazard types, and designing two targeted questionnaires.
- Led a seven-person team to conduct on-site surveys, interviews, inspections, and drone photography, collecting data on traffic flow, vehicle management, pedestrian movement, and infrastructure.
- Utilised SPSS and LLM tools for analysis of survey data and interview recordings, uncovering key issues and proposing actionable solutions that provided insights into the city's traffic safety and planning.
- Authored two technical documents: one analysing the relationship between traffic safety behaviour, attitude, and cognition using Structural Equation Modeling; the other assessing Kaiping City's traffic governance through Grey Relational Analysis and Fuzzy Evaluation. The project's final report was subsequently adopted by the Municipal Transportation Bureau.

INTERNSHIP

Guangzhou Xueshujia Software Technology Co., Ltd. **01/2025-05/2025**

Simulation Assistant Engineer

- Deployed a multi-agent reinforcement learning framework for CAV control within the company's traffic simulation software, to dissipate shock waves on smart highways.
- Benchmarked the software against leading simulation platforms like SUMO and CityFlow, and raised optimisation suggestions to enhance product competitiveness.
- Researched a wide range of traffic-related competitions and conferences, analysing the technical trends in intelligent transportation, vehicle-infrastructure cooperation, and big data applications, providing strategic insights for industry-academia collaboration.

EXTRACURRICULAR ACTIVITIES

Innovation and Entrepreneurship Incubation Park **03/2024-03/2025**

AIGC Imaging and Media Entrepreneurial Project | Founder and Technical Lead

- Designed customized image generation workflows based on Stable Diffusion and Midjourney, providing AIGC solutions for industrial clients such as shipping logistics and sports apparel.
- Developed special-effect process for e-commerce model clothing replacement, product rendering, and cosplay photo effects, estimated to cut costs by ~20% and boost efficiency by ~30%.

Voluntary Teaching Programme at Maoming Country Primary School **07/2023**

- Designed and delivered interactive lessons for primary school students, while engaging with local communities to gather insights into rural revitalisation and industrial development.
- Wrote a community report and four promotional articles to publicise local industries and rural life, inspiring more students to participate in volunteer education initiatives.

AWARDS & HONOURS

- Meritorious Winner (M Award), Interdisciplinary Contest in Modeling (ICM), COMAP. **2025**
- Best Poster Award, 25th COTA International Conference of Transportation Professionals. **2025**
- Third Prize, China Undergraduate Mathematical Contest in Modeling, Guangdong Division. **2024**
- Second Prize, 19th College Students Transportation Science and Technology Competition. **2024**
- Third Prize, 18th "Challenge Cup" College Students' Extracurricular Academic Science and Technology Works Competition, Guangdong Division. **2024**
- Science and Technology Innovation Model Scholarship, Guangzhou Maritime University. **2024**
- Merit Scholarship (Third Class), Guangzhou Maritime University. **2023**

SKILLS & LANGUAGES

- **Technical Tools:** Python, MATLAB, SUMO, LaTeX, Git, SPSS, Photoshop, Illustrator
- **Languages:** English (fluent), Mandarin (native), Cantonese (native)